The Customer Value Story: A Predictive Segmentation Journey

1. Introduction & Data Connection

Every customer tells a story through their data. To read these stories, we established a robust data pipeline connecting Python to a MySQL 'superstor_db'. By joining customer dimensions with sales facts, we consolidated thousands of transactions into a single view of customer behavior.

2. The Quest: Identifying High-Value Customers

In a sea of thousands of orders, not all transactions contribute equally to business health. Our goal was to build a Classification Predictive Model to segment customers into 'High' and 'Low' value groups based on their profitability.

Using the median profit as a threshold, we defined 'High-Value' customers as those whose total profitability meets or exceeds the middle benchmark of our database.

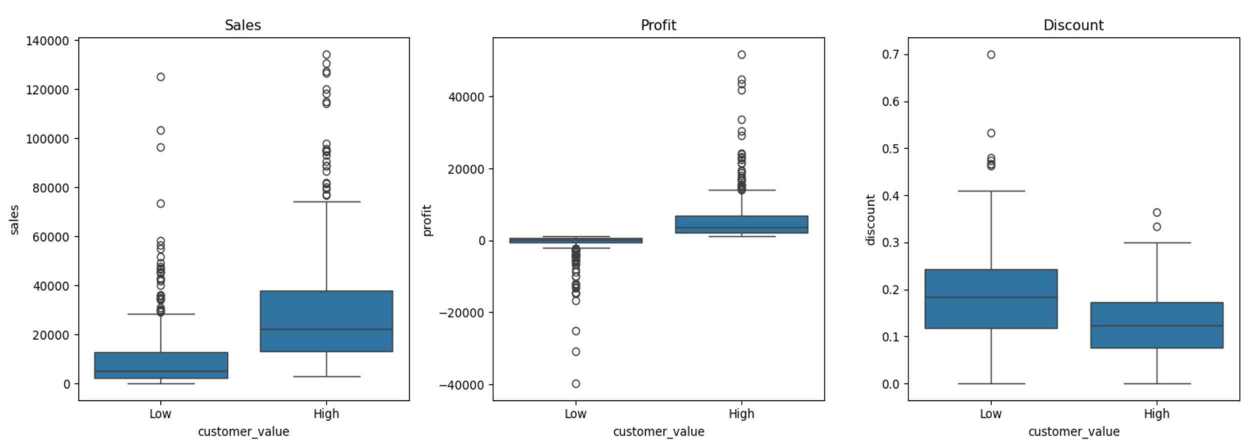
3. The Patterns in the Data

Before building the model, we observed clear behavioral differences between the groups:

- Sales: Significantly higher for High-Value customers.

- Profit: Deeply negative for many Low-Value customers, indicating losses.

- Discounts: Noticeably higher in the Low-Value segment, suggesting that heavy discounting may be eroding margins.

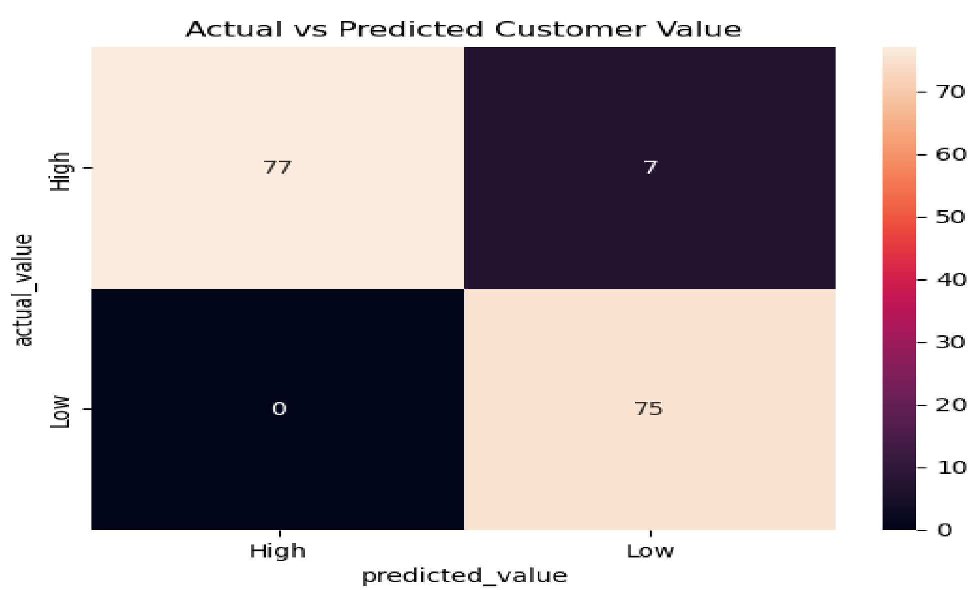


4. Model 1: The Full Feature Approach

We first trained a Logistic Regression model using all primary features: Sales, Quantity, Discount, Profit, and Order Frequency. The results were exceptional.

Model Accuracy: 95.6%

The model showed high reliability, successfully identifying nearly all High-Value customers with zero false positives.



5. Model 2: The Pure Behavior Approach

After completing the profit prediction analysis, an additional business challenge was explored:

"Can we predict customer value using only customer behavior without directly using Sales or Profit values?"

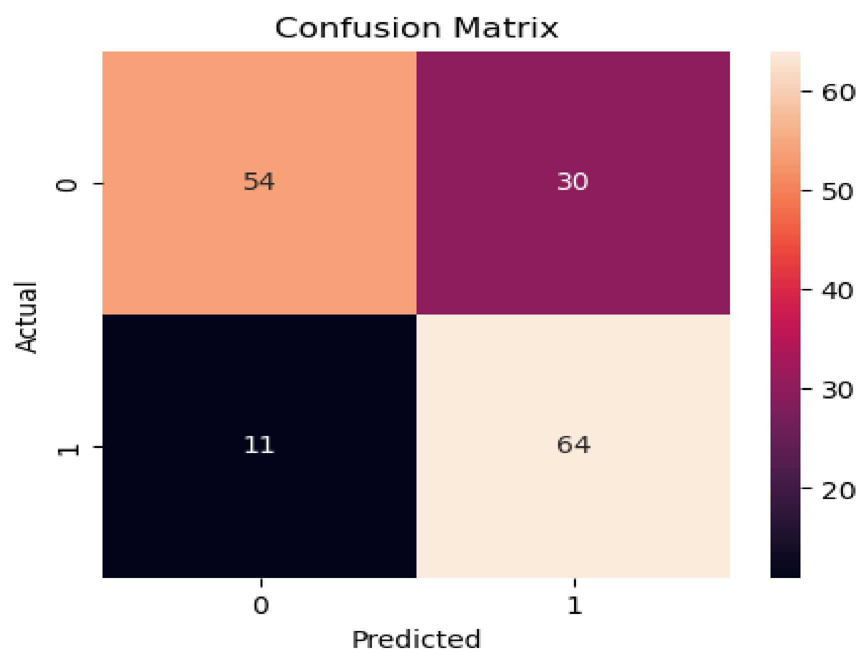
The purpose of this challenge was to understand whether customer purchasing behavior alone could help identify valuable customers.

Instead of depending on financial variables like:

- Sales
- Profit

the model focused only on behavioral indicators such as:

- Purchase Quantity
- Discount Usage
- Purchase Frequency



Model Accuracy: 74%

The model achieved an accuracy of 74.2%, which shows that customer behavior contains meaningful patterns that can help estimate customer value.

Business Interpretation

The results suggest that:

- Customers who purchase more frequently tend to have higher business value
- Discount behavior influences customer purchasing patterns
- Quantity and repeat purchases are important indicators of customer engagement
- The model could not fully capture profitability Some high-frequency customers may still generate low profit due to excessive discounts

6. Strategic Insights

- High-Value customers are characterized by higher sales and controlled discount usage.
- Low-Value customers often depend on heavy discounts, which do not necessarily improve the company's bottom line.
- A 95.6% accurate model allows the business to proactively identify which customers to target with loyalty programs.

7. Actionable Recommendations

- Focus Retention Efforts: Direct loyalty programs toward predicted High-Value customers to maximize ROI.
- Monitor Discounting: Carefully review discount-heavy strategies for Low-Value segments to avoid turning profit into loss.
- Targeted Marketing: Use the model to segment mailing lists, ensuring expensive marketing campaigns aren't wasted on low-margin segments.
- Continuous Improvement: Enhance the model by adding 'Recency' and 'Tenure' data to better predict long-term customer lifetime value.

8. Conclusion

By applying predictive classification, we transitioned from simply looking at past sales to predicting future customer potential. The models developed provide a reliable foundation for data-driven decision-making, helping the business balance sales volume with true profitability.